

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_random_replay.
- Best TSPLIB holdout gap: tsplib_random_replay.
- Best synthetic holdout gap: tsplib_random_replay.

Run Metadata

- run_name: run_20260513_154304_s
- started_at_local: 2026-05-13 15:43:04
- finished_at_local: 2026-05-13 15:52:48
- duration_hhmm: 00:10
- duration_seconds: 584.328
- seed_offset: 18000
- replicate_label: s
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.833514	0.56	-0.02988
tsplib_random_replay	random	score_only	False	0.13438	0.0	0.085515	0.0	0.76	0.0
tsplib_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.903635	0.56	-0.027561
tsplib_failure_replay_compression	failure	novelty_gate	True	0.213387	0.064916	0.159398	0.0	0.56	0.0

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 3, mean accepted code novelty 0.833514, and final complexity 0.56.
- Adaptation efficiency -0.02988 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.13438, synthetic 0.0, combined 0.085515.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.13438 across 7 instances; family means: ch=0.139876, kroD=0.13276, pcb=0.212887, pr=0.134478, rd=0.133375, st=0.047407.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3358, "family": "a", "name": "a280", "optimality_gap": 0.302055}, {"best_known_cost": 14379, "cost": 18059, "family": "lin", "name": "lin105", "optimality_gap": 0.255929}, {"best_known_cost": 7542, "cost": 8521, "family": "berlin", "name": "berlin52", "optimality_gap": 0.129806}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.903635, and final complexity 0.56.
- Adaptation efficiency -0.027561 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Suite Results

Replay Condition	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (mean)	Complexity	Replay Mode
tsplib_no_replay	0.2134	0.0649	0.1594	0.8335	0.56	none
tsplib_random_replay	**0.1344**	**0.0**	**0.0855**	0.0	0.76	random
tsplib_failure_replay	0.2134	0.0649	0.1594	0.9036	0.56	failure
tsplib_failure_replay_compression	0.2134	0.0649	0.1594	0.0	0.56	failure + compression

Interpretation

- ****Transfer and Optimality:****
- The ****random replay condition (tsplib_random_replay)**** achieves the best transfer performance both on held-out TSPLIB instances and synthetic holdout sets:
 - Lowest mean TSPLIB gap: 0.1344 (compared to ~0.2134 for others).
 - Lowest mean synthetic gap: 0.0 (others ~0.0649).
 - Lowest mean transfer gap: 0.0855 (others ~0.1594).
- This indicates superior transfer capability under random replay compared to no replay, failure replay, and failure replay with compression.
- ****Code Novelty vs. Transfer:****
- Random replay shows ****zero code novelty**** (mean_code_novelty = 0.0), substantially lower than no replay (0.8335) and failure replay (~0.9).
- Despite lower code novelty, random replay improves transfer gaps markedly.
- This suggests ****lexical novelty reduction**** (less code change) is associated here with better transfer performance, reflecting that improvements are not due to new code inventions but more effective reuse (random replay benefits).
- ****Failure Replay Conditions:****
- Both failure replay variants (with and without compression pressure) perform identically on transfer metrics and optimality gaps, matching no replay, with higher code novelty (~0.9 for failure replay vs 0 for compression variant).
- Compression pressure reduces code novelty but does not improve transfer or optimality gaps relative to no replay failure replay.

- ****No Replay Condition:****
- Baseline optimality and transfer gaps are higher, and code novelty is high (~0.83), indicating more novel code is produced but with poorer transfer outcomes compared to random replay.
- ****Behavior Profiles & Complexity:****
- Random replay exhibits a more complex final behavior profile (complexity = 0.76) versus others (~0.56).
- No replay and failure replay conditions converge to balanced behavior profiles with lower complexity.

Key Takeaways

- ****Random replay yields best transfer and optimality metrics despite having zero code novelty****, indicating the improvements come from effective replay rather than algorithmic novelty.
- ****High code novelty in no replay and failure replay does not translate to better transfer or held-out performance.****
- ****Failure replay with compression reduces code novelty but does not improve transfer or optimality gaps compared to failure replay without compression, indicating compression pressure alone does not enhance transfer.****
- ****Distinguishing replay modes is critical: random replay significantly outperforms no replay and failure replay on transfer, despite simpler (less novel) solutions.****

****Conservative Conclusion:****

The benchmark suite results confirm that random replay condition provides superior transfer capability and lower optimality gaps on both TSPLIB and synthetic holdout data. This advantage occurs alongside a pronounced reduction in code novelty, suggesting that in this context, transfer improvements stem from replay mechanism efficacy rather than novel code creation. Failure replay with or without compression yields no transfer gains relative to no replay, despite higher code novelty.